

Finding Influencers in Complex Networks: A Novel Method Based on Information Theory

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Abstract—Key node identification is one of the most fundamental and significant problems in complex network research. This article presents a novel approach called information-theory-based centrality C_H to distinguish influential nodes in complex networks. From an information-theoretic perspective, nodes involved in many information channels will improve the quality of communication. Thus, the greater the number of information channels a node participates, the more the importance of this node during information transmission. With this motivation, the local and global connectivity are measured according to the resistance distance, which captures the characteristic that multiple available routes reduce the resistance of information transmission on networks. We further model the probability of information diffusion according to the local and global connectivity and put forward an information-theory-based centrality to identify influential nodes. Finally, extensive evaluations are implemented on real-world networks to verify the feasibility and effectiveness of the proposed method. According to the results, our proposed method outperforms classical methods in identifying influential nodes and also indicates the potential for analyzing the influence evolution of networks, which shows a positive and effective impact on locating influencers and predicting potential key players in the future.

Index Terms—Complex networks, influential node identification, information theory, node centrality, resistance distance.

I. INTRODUCTION

THE last decade has witnessed the rise of a new interdisciplinary field of network science, which has attracted interest from a wide range of disciplines spanning sociology, physics, medicine, biology, and economics [1]–[3]. In the real world, a large number of complex systems, such as social systems, biological systems, infrastructural systems, and technological systems, can be modeled as large-scale complex networks with objects as vertices and associations between objects as edges [4]–[6].

The identification of influential nodes has always been a hot topic of network science [7]–[9]. The ability to efficiently and effectively recognize the crucial nodes in complex networks will help us to better manage the explosion of an epidemic, optimize the utilization of restricted resources to promote information

proliferation, undertake precision marketing for e-commercial networks, prevent blackouts in power grids, identify potential drug target candidates in protein networks, and design strategies to prevent communication collapses in telecommunication networks [10]–[14].

Many scholars have conducted in-depth research on the subject of identifying the influential nodes of complex networks and proposed a series of metrics [7], [15]. The structural-center-based indexes include the vertex connectivity, expansion parameter, tenacity, isoperimetric number, roughness, scattering number, natural connectivity, fault diameter, and algebraic connectivity. These indexes are calculated based on the node topological centrality without taking advantage of other information [16]–[20]. Among the systematic analysis methods, the most commonly used method is the node deletion method [21], [22], which identifies influential nodes by exploring the influence on the network after deleting the node. However, most of the previous methods in complex network analysis fail to evaluate a node's importance from the perspective of the roles of nodes in dynamical processes. Recent works shed lights on the dynamic propagation method [23], [24]. For example, Yang *et al.* [24] proposed a weighted degree decomposition approach to identify the most influential intruders in a dynamically evolving community.

Motivating application: Without loss of generality, information, viruses, and traffic in networks during dynamical processes are all referred to as network flow. In complex networks, the information flow and propagation is one of the core functions of complex networks [25]. Information is mainly transmitted through the interconnection of network nodes.

Take scientific collaboration network, one of the most common complex networks, as an example. Scientists cooperate with each other to form a collaborative network with scientists as nodes and partnership as edges. In scientific collaboration networks, links are established through the collaboration of scientists, and ideas flow and propagate through cooperation [26]–[28].

Fig. 1 shows the subgraph of the collaboration communities of Prof. Jiawei Han and Philip S. Yu on the topic *Data Mining*. In this figure, one node is an author and a weighted edge indicates the collaboration of two authors. The weight of an edge is proportional to the number of coauthored papers between two authors. No edge exists between two nodes in the case of no coauthored paper, and correspondingly, the weight is zero.

Jiawei Han and Philip S. Yu are among the most frequently cited scientists, as shown in Fig. 4 in Section IV,

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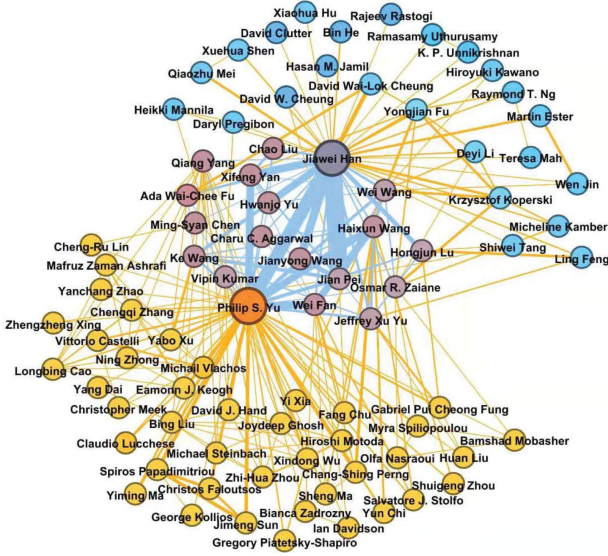


Fig. 1. Subgraphs of the collaboration communities of Prof. Jiawei Han (the large purple node) and Philip S. Yu (the large orange node) in the research area *Data Mining*. The rosybrown nodes in the middle part bridge the two communities by connecting both Prof. Jiawei Han and Philip S. Yu in light blue edges with papers coauthored by both scientists. The blue nodes in the upper community on the right represent coauthors collaborating with Prof. Jiawei Han, and the yellow nodes in the bottom community on the left represent coauthors collaborating with Prof. Philip S. Yu. The edges are weighted according to the number of coauthored papers between two nodes. For example, the bold light blue edges between Prof. Jiawei Han and Philip S. Yu and their common coauthors reflect more coauthored papers than the light orange edges within a community.

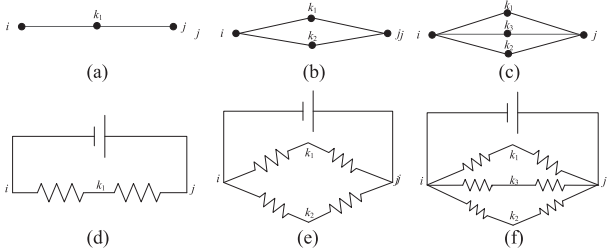


Fig. 2. Three graphs and their corresponding resistor networks. (a) G_1 . (b) G_2 . (c) G_3 . (d) $N(G_1)$. (e) $N(G_2)$. (f) $N(G_3)$.

TABLE I
EXAMPLES OF THE CITATIONS OF THE BRIDGE NODES IN FIG. 1

coauthors	#Citations
Jian Pei	95447
Vipin Kumar	92279
Qiang Yang	55028
Charu C. Aggarwal	50658

and their coauthors form two communities through tight inner communications. Note that structural holes [29] exist between the two parts of the network. Among published scientists, authors coauthoring with both Jiawei Han and Philip Yu span structural holes with their multiple paths and connect the communities by serving as bridges. Moreover, these authors contribute a large number of coauthored papers, as the weighted edges indicate. Most of these bridge nodes are influential scientists in the research area *data Mining*, as the citations in Table I indicate.

Scientists accumulate and propagate self-influence through knowledge exchange and collaboration. The knowledge flow and propagation in complex networks are closely related to the distance and connectivity between nodes: the more connections and the shorter distance there are between individuals, the greater the possibility of information propagation will be. However, traditional centrality measures fail to characterize these features for identifying influential spreaders in specific fields.

Our proposed work is trying to identify influential scholars in scientific research, which can be characterized as an influential node identification problem in complex networks.

Challenges and contributions: In light of this, this article takes full considerations of the information diffusion and propagation dynamics in complex networks and presents an approach called information-theory-based centrality to identify influential nodes in complex networks. The main contributions of this article can be summarized as follows.

- 1) Observing the fundamental role of multiple paths in facilitating information diffusion and interactions between nodes, we model the local and global connectivity of nodes based on the underlying structures of networks, respectively.
- 2) Modeling information diffusion through nodes along multiple channels, we propose a novel framework to identify influential nodes from the perspective of information theory.
- 3) To test the practicability and effectiveness of our proposed centrality, experiments are implemented on real-world networks. The results show that our proposed method outperforms classical methods and further uncovers potential patterns of the influence evolution.

The rest of this article is designed as follows. In Section II, this article analyzes the influential node identification methods. Detailed descriptions of the local and global connectivity, and the information-theory-based centrality, C_H for short, are presented in Section III. In Section IV, experiments are implemented on real-world networks and results are analyzed. Finally, Section V concludes this article.

II. RELATED WORK

Influential node identification, also known as key node analysis in complex networks [7], [15], can be roughly divided into the following categories.

- 1) *Local topological information method:* A simple method to evaluate influential nodes is to count how many neighbors it is linked to, namely, to determine its degree. Previous studies show that large-degree nodes matter in sustaining network connectivity, enhancing transport capacity and preventing infectious disease propagation. Chen *et al.* [16] suggested a semilocal centrality indicator to estimate the importance of nodes in a network. Kitsak *et al.* [17] claimed that the position of a node is more important than the number of its associated neighbors and presented a K-shell decomposition method to strip the external nodes layer by layer.

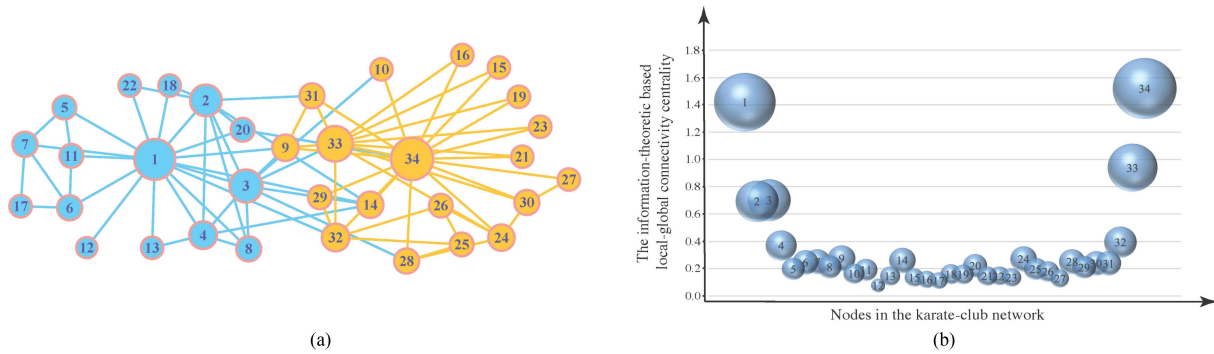


Fig. 3. (a) Karate-club network. (b) Ranking of nodes in the karate-club network based on our information-theory-based centrality C_H .

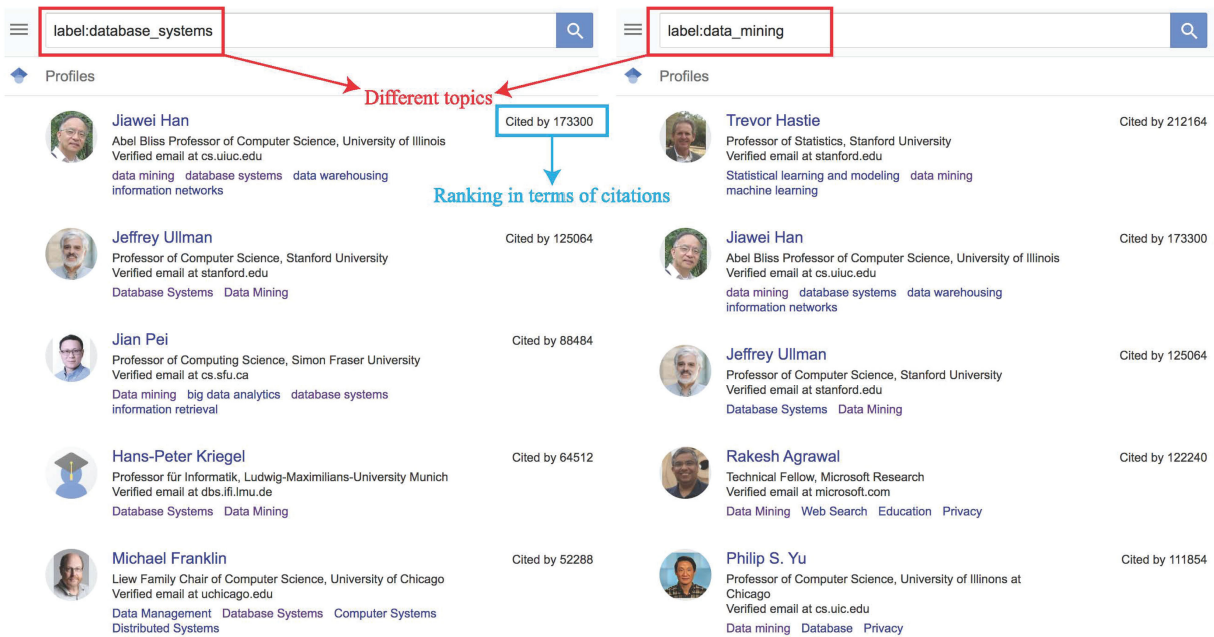


Fig. 4. Scholar rankings varying on the different topics *database systems* and *data mining* in terms of Google Scholar citations.

- 2) *Path-based methods*: In transportation, communication, and other networks, there exist some nodes that are connected to several core areas in the network. These nodes often serve as a bridge and are responsible for the information transmission between several modules in the network [18], [30]–[33]. The most commonly used path-based importance measure is betweenness. Betweenness centrality is a measure to what extent a node lies on the shortest paths between other nodes. Nodes with high betweenness may be influential in a network by assuming that information primarily flows over the shortest paths. Besides, Freeman [30] calculated the average distance from the target node to all other nodes in the network to measure node importance. By taking into account the total number of walks between a pair of nodes, Katz [32] put forward Katz centrality to compute the relative degree of influence of a node within a social network. Stephenson and Zelen [31] believed that information propagation on the network edge will cause information loss and presented an information indicator to measure node impor-

tance. Estrada and Hatano [18] suggested a measure of the communicability of a complex network, considering all different length path information between node pairs by assigning different weights to different length paths. Meo *et al.* [33] introduced the concept of node navigability, interpreted as the property that can be reached from anywhere else by a short walk, and successfully applied it to large-scale networks.

- 3) *Systematic analysis methods*: The most commonly used systematic analysis method is the node deletion method, which measures the importance of the node by exploring the impact on the network after deleting the node. Many scholars have conducted related research, for example, Tan *et al.* [21] presented a method of node contraction to identify key nodes in the network; Ren *et al.* [22] put forward a stochastic model and measured the cascading failure caused by node removal to identify influential nodes in a communication network.
- 4) *Dynamic propagation method*: In complex networks such as epidemic networks [34], rumor networks [23], and

social networks [35], [36], where there is information diffusion, the dynamical characteristics of the networks are also key factors in identifying important nodes of the networks. For example, Doostmohammadian *et al.* [37] used the centrality of nodes to control target groups in a community to effectively inhibit the spread of infectious diseases; Yang *et al.* [24] proposed a weighted degree decomposition approach to identify the most influential intruders in a dynamically evolving community. In terms of practical results, the methods based on network information propagation ability couple the structural information of the networks and the dynamics features, thus enabling more accurate identification of influential nodes.

III. INFORMATION-THEORY-BASED FRAMEWORK FOR FINDING INFLUENCER

In this section, we propose the local and global connectivity to quantify how a node participates into information diffusion through multiple information channels and further present a novel centrality to evaluate node influence in terms of information theory.

A. Characterizing the Multiple-Route Distance Diminishment Property

Consider a connected network $G = (V, E, W)$, where V is the vertex set with n nodes and E is the set of edges. The weight set W consists of the positive edge weight w_{ij} , which indicates the strength of the connection between node i and node j .

The connectivity of nodes is vital to information diffusion. Information is considered more likely to propagate between nodes with multiple paths. Fig. 2(a)–(c) gives three examples of interconnected nodes. In contrast to node i and node j in Fig. 2(a) and (b), information more easily reaches node j from node i in Fig. 2(c) because there are more paths, and *vice versa*.

The “multiple-route distance diminishment” feature is fundamental in dynamical processes in complex networks. That is, when multiple paths exist between two nodes, the distance between the two nodes decreases, especially in social scenarios under a great variety of circumstances.

However, traditional measures fail to capture the feature. For example, the shortest distances between node i and node j in Fig. 2(a)–(c) are all equal to 2. In comparison with the shortest distance, the resistance distance can be used to reflect the “multiple-route distance diminishment” feature.

Electrical network analysis views a graph as an electrical network with resistors as edges. Correspondingly, the resistance distance between two nodes of a graph is equal to the effective resistance between the respective two points of the resistor network.

Inspired by this, we associate a connected network with a resistor network by replacing each edge of the network with a resistor. More specifically, the resistor network $N(G)$ of G has a node for each vertex in V , and there is a resistor between the corresponding nodes in $N(G)$ for every edge in E .

Fig. 2(d)–(f) gives the corresponding resistor networks of the three original graphs in Fig. 2(a)–(c), respectively. For simplicity, the weights of the edges are equal to 1, and the corresponding resistances are units of 1.

The resistance distance between two nodes of a graph G is equal to the effective resistance between the respective nodes of the resistor network $N(G)$. That is, given node i and node j , the resistance distance r_{ij} is the effective resistance between i and j .

As shown in Fig. 2, with each edge being displaced by a unit resistance, the resistance distances between node i and node j are 2, 1, and 2/3 for G_1 , G_2 , and G_3 , respectively.

The resistance distance will decrease if more routes are available among nodes, and it will increase if these routes are disconnected. For example, in Fig. 2(f), the resistance distance decreases to 2/3 when a route through k_3 is added between node i and node j in G_2 . Comparatively, in Fig. 2(e), the resistance distance increases to 2 when one removes node k_2 on the path in G_2 . It is easy to verify that the resistance distance is nonnegative, symmetry and triangular inequalities.

B. Information-Theory-Based Centrality

Nodes in multiple paths facilitate the network connectivity and information diffusion. Regarding paths connecting nodes as “channels,” information diffusion is analogous to information that reaches the target node from the source node through “channels” in information theory. From the perspective of information theory, nodes with many paths contain richer information than nodes with fewer paths and greatly enhance the network communication capability through informative channels.

Inspired by this knowledge, we propose information-theory-based centrality to model the roles of nodes during dynamic processes. In terms of the resistance distance, we first define the local connectivity C_{ij} of node i and node j as follows:

$$C_{ij} = 1/r_{ij} \quad (1)$$

where r_{ij} is the resistance distance between node i and node j .

The local connectivity quantifies the desirable characteristic of multiple routes facilitating the information diffusion between nodes. That is, multiple paths reduce the resistance distance and ensure stronger connectivity between node i and node j , which facilitates the information transmission between the nodes.

Through the entire network, a node can reach a set of nodes to which it is connected with paths. Correspondingly, the global connectivity of nodes is further proposed to measure to what extent a node participates in information transmission through the entire network.

The global connectivity metric C_i of node i in G is defined as follows:

$$C_i = 1 / \sum_j r_{ij} = 1/r_i \quad (2)$$

where r_{ij} is the resistance distance between node i and node j , and r_i is the sum of the resistance distances between node i and each connected node j in G .

By accounting for the underlying structures of networks, strongly connected nodes with multiple paths tend to both achieve higher global connectivity than weakly connected nodes and play key roles by participating in as many information channels as possible. Thus, the global connectivity metric indicates the potential of one node to influence other nodes across the network.

For example, in Fig. 1, multiple paths exist between Profs. Jiawei Han and Philip S. Yu through their common coauthors, and thus, the resistance distance decreases due to the stronger local connectivity between them according to (1). Furthermore, Prof. Jiawei Han collaborated with all of the authors in his community by interconnected paths, and the global connectivity further increases according to (2). Similar results hold for Prof. Philip S. Yu according to the local and global connectivity.

Based on the measures of the local and global connectivity, we propose the information-theory-based centrality of nodes as follows:

$$C_H(i) = - \sum_{i < j} \frac{C_i}{C_{ij}} \log \frac{C_i}{C_{ij}} = - \sum_{i < j} \frac{r_{ij}}{r_i} \log \frac{r_{ij}}{r_i}. \quad (3)$$

The information-theory-based centrality can be considered as a model of information diffusion. Assume that node i emits some information that flows through a network at random. The probability that the information reaches node j depends on the local connectivity of node i to node j . Information recursively propagates by random walk diffusion. Information diffusion depends on the global connectivity of node i to the other nodes in the network. A node can have a high rank if the information reaches as many nodes as possible. If a node is connected to other nodes through more paths, it will communicate with and influence other nodes more easily.

It is easy to verify that $\frac{r_{ij}}{r_i}$ is nonnegative and $\sum_j \frac{r_{ij}}{r_i}$ equals 1. The variable $\frac{r_{ij}}{r_i}$ is proportional to the probability of information diffusing from node j to node i among all of the nodes connected with node i , where multiple paths increase the odds of information diffusion between nodes. The centrality $C_H(i)$ quantifies the expected value of the amount of information transmitted through node i on a network.

The centrality $C_H(i)$ provides the framework to evaluate the influential nodes in terms of the roles of nodes during dynamic processes with the following advantages. First, the centrality uses the resistance distance to measure the connectivity with consideration of the “multiple-route distance diminishment” property. Second, the centrality utilizes the connectivity to measure the ability of nodes to play roles in dynamic processes. The higher the centrality is, the more influential node i is.

C. Analysis and Discussion

To rank nodes in terms of the metric C_H , the key is to calculate resistance distance between two nodes in the corresponding electrical network of a graph. Given a graph G , resistance distances between nodes can be computed according to the Moore–Penrose inverse of the Laplacian matrix of G .

The Laplacian matrix is a matrix representation of a graph. More specifically, the Laplacian matrix L of G is of the form

$L = D - A$, where D is the diagonal matrix of G and A is the weighted adjacency matrix. The element of the adjacency matrix A is $w_{ij} \in \{1, 2, 3, \dots\}$ if node (author) i coauthored one, two, three, or more papers with node (author) j correspondingly, and $w_{ij} = 0$, otherwise. The weights are normalized during the computation. The i th element D_{ii} of the diagonal of D is equal to $D_{ii} = \sum_{j \in \{1, \dots, n\}} w_{ij}$.

The Moore–Penrose inverse L^+ of the Laplacian matrix L , also known as pseudoinverse, is the generalization of the inverse matrix, which can be computed using the singular value decomposition. Indeed, the pseudoinverse L^+ of the matrix L with singular value decomposition $L = U\Sigma V^*$ is

$$L^+ = V\Sigma^+U^* \quad (4)$$

where Σ^+ is the pseudoinverse of Σ , which is formed by replacing every nonzero diagonal entry by its reciprocal and transposing the resulting matrix.

According to the generalized inverse L^+ of the Laplacian matrix L of G , the resistance distance r_{ij} between node i and node j is computed as follows:

$$r_{ij} = (L^+)_{ii} + (L^+)_{jj} - 2(L^+)_{ij} \quad (5)$$

where $(L^+)_{ii}$ and $(L^+)_{jj}$ are the i th and j th elements on the diagonal of the generalized inverse L^+ , respectively. The element $(L^+)_{ij}$ is on the i th row and the j th column of the matrix L^+ .

Iterative methods are widely used to solve the Moore–Penrose inverse of a matrix [38], [39]. This article adopts the Cholesky decomposition to compute the inverse of a large-scale sparse matrix. All of the time-consuming parts are parallelized and roughly linear time in the distributed computing environment.

More specifically, given an $m \times n$ matrix L with $m \geq n$, parallel algorithms [40]–[42] are developed to compute Moore–Penrose inverse by implementing Cholesky decomposition and the inversion of a symmetric positive-definite matrix M derived from L . The time complexity for the Cholesky decomposition could reduce to $O(n)$, while the time complexity for the inversion of the matrix M could reduce to $O(\log r)$, where r is the rank of M .

Identifying influential nodes in complex networks is a challenging process. We realize that our study has limitations. For example, our method mainly evaluates the importance of nodes from the perspective of information diffusion and propagation dynamics based on topological structures. More deliberated factors will be considered to meet the real applications, such as systematic functions of networks. Moreover, our method cannot be directly applied to directed graphs, since the incoming and outgoing edges of each vertex are counted respectively. In the follow-up research, the extension to the above limitations will be further investigated.

IV. EXPERIMENTS

Experiments are implemented in this section to evaluate the effectiveness and efficiency of our proposed method over a wide range of centrality measures.



Fig. 5. Comparisons of evolution of the number of citations *per year* between the *top-1* scholars of our proposed method and the baseline methods. The subplots (a)–(d) exhibit how citations of the *top-1* scholars change *per year* on the topics, respectively. Each bar corresponds to the citation of the *top-1* scholar during the specific year. (a) Evolution of the number of citations per year of the *top-1* scholar on the topic *database systems*. (b) Evolution of the number of citations per year of the *top-1* scholar on the topic *machine learning*. (c) Evolution of the number of citations per year of the *top-1* scholar on the topic *Bayesian networks*. (d) Evolution of the number of citations per year of the *top-1* scholar on the topic *web mining*.

A. Case Study

First, we conduct our experiments on the dataset of a karate club in the real world [43] as a case study. Fig. 3 depicts the friendship network of 34 members in a karate club, where nodes represent the members and edges connect the pairs of people who are friends. Among these nodes, node 1 and 34 stand for the instructor and the club founder, respectively. With their extensive interconnections to other people, node 1 and 34 are central in the social network.

Because of the conflicts between the club administrator and the instructor, the club was split into two clubs, and each new club consisted of the individuals indicated by different colors in Fig. 3(a). At the boundaries of the two clubs, the interactions are few and structural holes [29] exist between the two parts of the network. Nodes spanning structural holes with their multiple paths, such as node 2, 3, 4, and 20 in the club on the left and node 9, 14, 32, and 33 in the club on the right, play key roles by offering informational advantages relative to the other nodes. According to the real-life scenario, the above nodes should be identified for their critical roles in information diffusion across the two parts of the network.

In consideration of the roles of the members in the karate club, we evaluate their centrality in terms of our proposed centrality. The results are shown in Fig 3(b), where the sizes of the nodes are linearly proportional to their centrality. As illustrated, node 1 and 34 are the most important ones in the network, which is consistent with their central roles in the network. Furthermore, node 2, 3, and 33 are also among the important ones, where nodes 3 and 33 maintain multiple connections to other members; in particular, they reach out to people in the other club. In addition, node 4, 9, 14, 20, 24, 28, and 32 stand out due to their multiple paths; in particular, they have contacts with people in different clubs rather than only in the same group.

B. Experiment Settings

To validate our methods, we further investigate the information-theory-based centrality in real-world networks.

1) *Topic-Oriented Academic Coauthor Networks*: In topic-oriented academic coauthor networks, individuals are linked if they have coauthored one or more papers. One paper covers one or more topics, and one author usually conducts research on

TABLE II
IDENTIFYING INFLUENTIAL SCHOLARS

Topic-based networks	top-k	C_D	C_C	C_B	C_{core}	C_I	Our method C_H
<i>Database Systems</i>	<i>top-1</i>	119496/136	124762/136	88364/136	124762/136	88364/136	166169/166
	<i>top-5</i>	42170/86	23159/62	20704/63	41032/87	20704/63	45539/90
	<i>top-10</i>	31375/72	28712/71	34170/78	44191/82	34170/78	47666/87
	<i>top-15</i>	39982/78	33887/74	27369/70	44738/81	27369/70	50300/86
	<i>top-20</i>	35398/74	36343/74	38485/76	38365/75	38485/76	43882/82
<i>Bayesian Networks</i>	<i>top-1</i>	79753/124	79753/124	79753/124	17178/61	79753/124	89108/137
	<i>top-5</i>	37958/79	22260/59	38207/81	12049/53	38537/79	64973/106
	<i>top-10</i>	23179/61	21439/56	42453/87	16010/53	23152/61	44031/83
	<i>top-15</i>	23674/62	26354/66	31707/73	17662/52	18029/53	33828/74
	<i>top-20</i>	25721/65	22634/60	32973/69	15343/50	16206/51	30986/71
<i>Web Mining</i>	<i>top-1</i>	40591/94	32493/103	32493/103	6648/47	44630/88	96705/108
	<i>top-5</i>	18990/53	16092/59	15408/57	3114/20	21625/60	36118/61
	<i>top-10</i>	23960/50	12984/50	20667/62	4917/29	15648/49	30568/61
	<i>top-15</i>	21101/48	10332/44	23154/55	8761/36	12432/41	26091/58
	<i>top-20</i>	18025/47	8819/39	21406/55	9650/39	12612/41	20190/49
<i>Machine Learning</i>	<i>top-1</i>	30628/84	154797/156	154797/156	104513/114	54363/82	285529/156
	<i>top-5</i>	11975/40	16531/49	12518/43	45226/72	7043/28	30420/56
	<i>top-10</i>	10900/38	13661/45	13980/44	26324/50	16045/45	33264/59
	<i>top-15</i>	12368/42	25427/55	23618/53	20317/46	13600/39	50757/63
	<i>top-20</i>	14993/45	20991/51	20380/50	17534/44	13042/40	44031/63

multiple topics. For an individual scholar, his or her influence varies according to the different topics related to his or her research. For example, Professor Jiawei Han, an eminent computer scientist, conducts research on a variety of topics, including *data mining*, *database systems*, *data warehousing*, and *information networks*. Fig. 4 shows the influence ranking of scholars on the topics *database systems* and *data mining*, where Prof. Jiawei Han ranks first and second, respectively, and the other most influential scholars vary on the two topics.

The experiments are conducted on topic-oriented academic coauthor networks extracted from ArnetMiner, which is a search system for academic publications on the Internet. The coauthor dataset is composed of 640 134 authors and 1 554 643 coauthor relations [44], and Tang *et al.* [45] proposed an Author–Conference–Topic model to extract the research topics for each author.

2) *Baseline Methods*: We compare our proposed approach (C_H for short) with classical centrality measures: degree centrality (C_D), closeness centrality (C_C), betweenness centrality (C_B), coreness centrality (C_{core}), and information centrality (C_I) by identifying influential scholars on a variety of topics *database systems*, *machine learning*, *web mining*, and *Bayesian networks*.

3) *Evaluation Metrics*: For assessing the influence of academic scholars in terms of all the methods, two typical evaluation metrics are used.

- 1) *Citations*: A citation is a reference to an article. The citation metric of an author indicates how many references the author gathers to his/her articles. The more citations one author accumulates, the more widely and frequently his/her articles are referenced. Thus, citations indicate how influential one author is.
- 2) *H-index*: Based on the set of a scientist's most cited papers, the H-index of an author is the highest number of his/her publications that received H or more citations each. The

H-index metric measures both the productivity and citation impact of the publications of an author.

For all of the authors contained in our dataset, we extracted their citation and H-index information from Google Scholar to evaluate their social influence.

C. Analysis of Experimental Results

1) *Identifying Influential Nodes*: The most influential scholars play critical roles in the development of scientific research. Specifically, finding an optimal set of key players in general graphs that optimize connectivity measures is typically NP-hard (NP refers to nondeterministic polynomial time) [46], which prohibits exact solutions of such problems for large-scale networks.

The advantages of our proposed method are validated in identifying both individual and a set of influential scholars with the highest numbers of citations and H-indexes, as shown in Table II.

For finding the most influential scholar ranked as *top-1*, our proposed centrality identifies Prof. Jiawei Han with the highest 166 169 citations and H-index of 166 on the topic *database systems* (see the first line in Table II). Prof. Jiawei Han is the most frequently cited author on the topic and accounts for the most vital individual node among all of the authors. In contrast, the classical centrality measures extract scholars with far fewer citations and lower H-indexes based on the structure properties such as the degree and the shortest paths.

A set of influential scientists are recommended correspondingly as *top-k* varies from *top-5* to *top-20*. To reflect the influence of a set of authors as a group, the average values of the citation and H-index metrics are computed to assess their performance.

Table II shows that our proposed centrality extracts a set of high-impact scholars with the highest average citations and H-indexes on the topic *database systems*. On the topic

machine learning (see *top-1* in the last line in Table II), our method outperforms all of the other methods by identifying Prof. Geoffrey E. Hinton, the outstanding scientist who won the ACM Turing award in 2019, as the *top-1* influential scholar with the highest citations 285 529 and the H-index of 156. Furthermore, *top-k* scientists with wider influence are located by our method with higher average citations and H-indexes. On the topics *web mining* and *Bayesian networks*, the advantages of our proposed methods are also observed by locating scholars with the highest (average) citations and H-index.

The above results suggest that our proposed method is more robust when the topics and the number of vital nodes vary. In contrast, the classical centrality measures fail to identify highly influential researchers, and their performance fluctuates heavily on various topic-oriented networks.

2) *Detecting the Influence Evolution of Individual Influential Scholars*: In real life, the influence of vital nodes varies over time under a great variety of circumstances. It is crucial but challenging to detect vital nodes with long-term influence in the related research fields. Hence, we further explore the evolution of the individual influential scholars identified by our method and the classical centrality measures.

Take the influence evolution of the *top-1* scholars as an example. For the *top-1* scholars extracted from the coauthor datasets up to 2009, Fig. 5 shows the comparisons of the evolution of the citations they receive *per year* on the selected topics from 2009 to 2018. The evolution of the H-indexes keeps similar trend. The data came from Google Scholar.

In contrast to the classical methods, the *top-1* scholars ranked by our proposed method maintained the highest citations each year during the following decade from 2009 to 2018 on all the topics. This result suggests that our method captures the fundamental factors for the evaluation of the node influence that stems from the underlying structures of interaction and collaboration.

Furthermore, on the topics *database systems* and *Bayesian networks* [see Fig. 5(a) and (c)], the *top-1* scholars identified by our method stand out among the other vital scholars identified by the classical methods, and their influence improves as the number of citations grows dramatically, while the results of the other methods change little. This pattern implies the rich-get-richer phenomenon during the evolution of academic influence.

On the topic *machine learning* [see Fig. 5(b)], although the *top-1* scholar identified by our proposed method, namely, Prof. Geoffrey E. Hinton, did not have the most citations among all of the vital scholars by the evaluated methods in 2009, the influence of Prof. Geoffrey E. Hinton has continued improving since 2015, as the citations have surged ahead. As indicated by the dramatically increasing number of citations in recent years, Prof. Geoffrey E. Hinton has gained a worldwide reputation for his work on artificial neural networks and deep learning and won the ACM Turing award in 2019. This distinction reveals that the burst in technological circumstances has reached a certain critical point.

On the topic *web mining* [see Fig. 5(d)], the *top-1* scholar ranked by our method has similar citations with other vital scholars highlighted by the classical methods in 2009. However, the gaps widened as the citations of the scholars identified by our

methods rose more quickly, while the citations of the scholars identified by the other methods declined.

The comparison suggests that our centrality identifies top scientists that maintain strong influence on the topics as time passes. Moreover, our centrality uncovers both typical and interesting trends during the evolution of social influence.

V. CONCLUSION AND FUTURE WORK

In this article, we have investigated a novel centrality in the framework of information theory, and it suggests a promising perspective to understand the organizing principles of complex networks.

Based on the observation of multiple-route distance diminishment, a system-level analysis is implemented to take advantage of both the local and global structure characteristics using the resistance distance. By modeling a network flow through local and global connectivity, it has been demonstrated that our proposed centrality has remarkable advantages over classical centrality measures in identifying both individual influential nodes and a set of vital nodes in a complex network.

Moreover, our method also indicates the potential for analyzing the influence evolution of networks with no prior knowledge, especially in considering the evolution of complex networks. Our results have strong potential significance for modeling network dynamics.

In the future, we will probe the extension of our method on directed graphs. Moreover, we will implement our metric for other domains of complex networks, such as Internet network and power grid network.

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